

Ozone Systems for Swimming Pools: Evidence-Based Research Report

Executive Summary

Ozone systems for swimming pools are secondary water treatment technologies that use ozone gas (O₃) to disinfect pool water and oxidize contaminants. Ozone is a blue-colored gas that's made up of three oxygen atoms. Ozone is useful in swimming pools because it effectively kills and inactivates bacteria, parasites, and viruses. It's also a good oxidizer, which means that it can chemically destroy contaminants in the water. These systems are gaining popularity as alternatives to heavy chlorine usage, offering reduced chemical exposure and elimination of chloramines. However, ozone leaves little or no residual in a pool or spa and is used in conjunction with an EPA registered primary sanitizer, meaning they cannot completely replace traditional sanitizers. While generally safer for swimmers than high-chlorine environments, ozone presents respiratory risks if exposure occurs in poorly ventilated areas.

Definition & Context

Ozone pool systems are water treatment technologies that generate ozone gas to disinfect swimming pool water. The use of ozone prevents the formation of chlorine or bromine containing byproducts, like chloramines and trihalomethanes. Ozone will even breakdown already formed unwanted chlorinated byproducts. These systems are commonly used in both residential and commercial pool settings as secondary disinfection systems.

NSF/ANSI Standard 50 requires that ozone be used in conjunction with chlorine. The systems typically cost between \$1,000 and \$3,000 to buy and install and are regulated as secondary treatment systems rather than primary sanitizers in most jurisdictions.

Benefits / Positive Effects

Benefit	Why It Matters	Supporting Evidence
Effective Pathogen Control	Kills bacteria, viruses, and parasites without harmful byproducts	Ozone effectively kills and inactivates bacteria, parasites, and viruses
Reduces Chemical Exposure	Minimizes swimmer contact with harsh chemicals	Ozone a much safer and healthier alternative to chlorine, which can produce toxic by-products such as chloramines
Eliminates Pool Odor	Destroys chloramines that cause "chlorine smell"	Ozone will even breakdown already formed unwanted chlorinated byproducts. This will reduce problems with red eyes and respiration problems
Reduces Eye/Skin Irritation	Eliminates chloramine formation that causes irritation	Ozone pools have no pool odor, and help reduce chemical exposure
Lower Chlorine Consumption	Reduces need for primary sanitizer chemicals	Ozone has the added benefit of destroying chloramines as well as reducing chlorine consumption
Easier Maintenance	Fewer chemical adjustments needed	Ozone pools are easier to maintain

Risks / Negative Effects

Risk	Potential Impact	Evidence / Source
Respiratory Health Effects	Moderate to Severe	Depending on the level of exposure, ozone can cause coughing and sore or scratchy throat, make it more difficult to breathe deeply and vigorously
Airway Inflammation	Moderate to Severe	Inflame and damage the airways. Make the lungs more susceptible to infection
Asthma Aggravation	Severe for susceptible individuals	Aggravate lung diseases such as asthma, emphysema, and chronic bronchitis. Increase the frequency of asthma attacks
High Installation Cost	Moderate (Financial)	Ozone pools have high upfront costs
Still Requires Chemicals	Mild (Operational)	They still require a chemical supplement
Safety Risks if Leaking	Moderate to Severe	can be dangerous
No Residual Protection	Moderate (Operational)	Ozone leaves little or no residual in a pool or spa

Toxicity & Safety Profile

Human Health Effects

Short-term Exposure: Even relatively low levels of ozone can cause health effects. Depending on the level of exposure, ozone can cause coughing and sore or scratchy throat, make it more difficult to breathe deeply and vigorously and cause pain when taking a deep breath.

Long-term Exposure: Long-term exposure to ozone is linked to aggravation of asthma, and is likely to be one of many causes of asthma development.

Exposure Thresholds

EPA's current National Ambient Air Quality Standards (NAAQS) are 0.070 parts per million (ppm), as the fourth-highest daily maximum 8-hour concentration, averaged across three consecutive years.

Health Agency Positions

- **EPA:** Regulates ozone as an air pollutant with established NAAQS
- **OSHA/EPA Pool Regulations:** Neither OSHA nor the EPA require these systems to capture or destroy the ozone. Some commercial systems that produce high amounts of ozone DO have extra safety equipment to prevent leaks, monitor ozone in the air, maximize the transfer into the water, and capture any escaped ozone.

Vulnerable Groups

People most at risk from breathing air containing ozone include people with asthma, children, older adults, and people who are active outdoors, especially outdoor workers. Children are at greatest risk from exposure to ozone because their lungs are still developing and they are more likely to be active outdoors when ozone levels are high.

Environmental Impact

Air Quality: Ozone in ambient air is a regulated pollutant. Pool ozone systems can contribute to indoor air ozone levels if not properly contained.

Water Impact: Ozone breaks down to oxygen in water, leaving no harmful residual chemicals.

Wildlife: Limited direct impact from pool systems when properly contained. Atmospheric ozone depletion has broader environmental consequences unrelated to pool systems.

Who Should / Should Not Be Exposed

Should Avoid or Use Extreme Caution:

- People with asthma, emphysema, or chronic bronchitis
- Children (especially those with respiratory conditions)
- Elderly individuals
- Outdoor workers in pool areas with ozone systems
- Anyone with compromised respiratory function

Generally Safe (with proper ventilation):

- Healthy adults swimming in properly maintained ozone-treated pools
- Pool users in well-ventilated outdoor areas
- Occasional swimmers without respiratory sensitivities

Safer Alternatives or Mitigation Strategies

Alternative Systems:

- UV disinfection systems
- Saltwater chlorination (chlorine generators)
- Advanced Oxidation Processes (AOP)
- Hydroxyl-based Disinfection Oxidation (HDO)

Mitigation Strategies:

- Ensure proper ventilation in indoor pool areas
- Install ozone leak detection and destruction systems
- Use ozone systems with automated shutoffs
- Regular air quality monitoring in enclosed spaces
- Proper system maintenance to prevent leaks

Comparison

Option	Pros	Cons	Best For
Ozone Systems	Effective disinfection, reduces chloramines, less chemical exposure	High cost, requires backup sanitizer, respiratory risks if leaked	Outdoor pools, users without respiratory issues
UV Systems	No chemical byproducts, effective pathogen control	No residual protection, high energy use	All pool types, especially indoor
Saltwater Chlorination	Gentler than traditional chlorine, familiar technology	Still produces chlorine, equipment corrosion issues	Residential pools, cost-conscious users
Traditional Chlorine	Low cost, proven effective, residual protection	Chloramine formation, skin/eye irritation, strong odor	Budget-conscious, simple maintenance needs

Sources

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